An Exact Equidistribution Law for the 2-adic Valuations of $3^r m - 1$

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Abstract

We prove that for all $r, k \geq 1$ there is *exactly one* odd residue m modulo 2^k with $\mathrm{v}_2(3^r m - 1) \geq k$. Consequently, for m drawn uniformly from the odd residues mod 2^k, $\Pr[\mathrm{v}_2(3^r m - 1) = k] = 2^{-k}$ — a geometric law with mean 2, *independent of* r. The result is an exact counting statement about the 2-adic valuation of $3^r m - 1$; it is the rigorous form of a folklore heuristic used in the study of $3x + 1$-type maps, but it is a statement about 2-adic arithmetic and does not depend on any conjecture. The theorem is formalised in Lean 4 and machine-checked with kernel-only axioms.

MSC: 11A07 (Congruences), 11S05 (2-adic fields), 60C05 (Combinatorial probability), 68V20 (Formalisation of mathematics).

1 Statement

Throughout, v_2 denotes the 2-adic valuation and m ranges over the odd residues modulo 2^k (there are 2^{k-1} of them).

Theorem 1.1 (Exact equidistribution)**.** *Let $r, k \geq 1$. There is exactly one odd $m \in [0, 2^k)$ with*

$$\mathrm{v}_2(3^r m - 1) \;\geq\; k.$$

Corollary 1.2 (Geometric law)**.** *Let m be uniform on the odd residues mod 2^k. Then for $1 \leq j < k$,*

$$\Pr\big[\mathrm{v}_2(3^r m - 1) = j\big] = 2^{-j},$$

independently of r; and $\Pr[\mathrm{v}_2 \geq k] = 2^{-(k-1)}$. Letting $k \to \infty$, the valuation $\mathrm{v}_2(3^r m - 1)$ is $\mathrm{Geom}(1/2)$ on $\{1, 2, \dots\}$, with mean $\mathbb{E}[\mathrm{v}_2] = \sum_{j \geq 1} j 2^{-j} = 2$.

2 Proof

Lemma 2.1 (Invertibility)**.** *3 is a unit in $\mathbb{Z}/2^k\mathbb{Z}$; hence 3^r is a unit and the map $m \mapsto 3^r m$ is a bijection of $(\mathbb{Z}/2^k\mathbb{Z})^\times$.*

Proof. $\gcd(3, 2) = 1$, so 3 is invertible modulo every power of 2; units form a group, so 3^r is a unit and multiplication by it permutes the units. □

Proof of Theorem 1.1. The condition $\mathrm{v}_2(3^r m - 1) \geq k$ is exactly $3^r m \equiv 1 \pmod{2^k}$, i.e. $m \equiv (3^r)^{-1} \pmod{2^k}$. By Lemma 2.1 the inverse $(3^r)^{-1}$ exists and is unique modulo 2^k, and it is odd (being a unit). Hence exactly one odd residue $m \in [0, 2^k)$ satisfies the condition. □

Proof of Corollary 1.2. Apply Theorem 1.1 at level j and at level $j+1$: the odd residues mod 2^k with $\mathrm{v}_2 \geq j$ are those lying in a single class mod 2^j, hence a fraction $2^{-(j-1)}$ of the odd residues; likewise $\Pr[\mathrm{v}_2 \geq j+1] = 2^{-j}$. Subtracting, $\Pr[\mathrm{v}_2 = j] = 2^{-(j-1)} - 2^{-j} = 2^{-j}$. The count is independent of r because Lemma 2.1 makes multiplication by 3^r a bijection of the units for every r. The mean is $\sum_{j\geq 1} j2^{-j} = 2$. □

Remark 2.2. Only two facts are used: that the base is *odd* (hence a 2-adic unit) and that the units form a group. So the theorem holds verbatim with 3 replaced by any odd a: for all $r, k \geq 1$ exactly one odd $m \in [0, 2^k)$ has $\mathrm{v}_2(a^r m - 1) \geq k$, and the induced valuation law is Geom(1/2) with mean 2, independent of a and r.

3 Formalisation

The statement and proof are formalised in Lean 4:

- `unique_inverse_mod` — uniqueness of the inverse of 3^r mod 2^k;
- `modular_solution_odd` — the solution is odd;
- `geometric_distribution_informal` — Theorem 1.1 itself, stated as the exact cardinality $|\{m < 2^k : m \text{ odd},\ k \leq \mathrm{v}_2(3^r m - 1)\}| = 1$. (The suffix is a historical misnomer: the statement is fully formal.)

It builds with `lake build CollatzProof.KodamaCollatz`, and `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]` for all three — the ambient kernel axioms only, no `sorry` and no custom axioms. The audit is kept as `CollatzProof/AxiomAudit.lean` so that it re-runs on every build. Corollary 1.2 is the elementary counting consequence and is not separately formalised.

4 Why it matters, and what it does not give

In the study of $3x+1$-type maps the accelerated step is $x \mapsto (3x+1)/2^{\mathrm{v}_2(3x+1)}$, and the heuristic "$\mathbb{E}[\mathrm{v}_2] = 2$, hence the expected factor per step is $3/4 < 1$" is standard. Corollary 1.2 is the exact statement behind that heuristic: the valuations really are Geom(1/2) when the argument is equidistributed over odd residues.

Remark 4.1 (Scope)**.** This is a statement about *uniformly distributed* residues. A single deterministic orbit is not a random sample: its valuations are fixed, not drawn. Consequently Corollary 1.2 does *not* imply that any particular orbit contracts, and it implies nothing about the Collatz conjecture. That a negative average drift need not decide a deterministic orbit is shown concretely by the divergent map $\lfloor \sqrt{2}\, n + 4 \rfloor$ of Althöfer–Zipproth [3].

Reproducibility

The counting statement is re-checked by `tools/verify_for_papers.py`; the Lean items build with `lake build`.

Acknowledgement

AI-assisted; every claim was verified by exact computation or Lean's kernel before inclusion.

References

[1] J. C. Lagarias, *The* $3x+1$ *problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.

[2] R. Terras, *A stopping time problem on the positive integers*, Acta Arith. **30** (1976), 241–252.

[3] I. Althöfer, T. Zipproth, *Divergence in a Collatz-type recursion*, preprint, August 2025, https://althofer.de/althoefer-zipproth-august-2025.pdf.

[4] The mathlib Community, *The Lean mathematical library*, CPP 2020.